Cogongrass Detection from Drone Imagery

Project Report: Models, Results, Findings & Roadmap

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Bottom line: a deployable tile-classification model detects cogongrass on a genuinely unseen drone collection at ~0.84 balanced accuracy (with AdaBN test-time adaptation). Architecture and preprocessing are exhausted; the remaining bottlenecks are DATA DIVERSITY and LABEL QUALITY, not the model.

1. Executive Summary

Goal: automatically detect the invasive grass cogongrass (Imperata cylindrica) in drone imagery. The project evolved from an initial detect-then-classify plan into a TILE CLASSIFICATION pipeline: cut each frame into a grid, classify each tile cogongrass / not, stitch into a coverage heatmap.

What works

- Deployable model: ResNet18 + domain-adaptation training + AdaBN ~= 0.84 balanced accuracy on a held-out collection (true cross-collection test).
- Reproducible pipeline: import -> tile -> train -> AdaBN inference -> heatmap.
- Honest evaluation: cross-collection (train one flight, test another) instead of a misleading random split that inflated scores to ~0.97.
- Test-time AdaBN is the single most effective technique (+2 pts, free, deployment-realistic).

The core finding

Every architecture and preprocessing lever we tried converges to ~0.82-0.84 cross-collection while in-collection stays ~0.97. That flat ceiling across two backbones (ResNet, DINOv2) and every knob proves the bottleneck is DATA, not the model. Separately, visual inspection showed most 'false positives' are unlabeled cogongrass -> the metrics are a floor limited by LABEL QUALITY.

2. Data

Close-up reference data (web)

ResNet50 trained on iNaturalist close-ups: cogongrass vs 10 confuser grass species. Test 91.9% - but it is a DIFFERENT domain (sharp close-ups) and does not transfer to oblique/aerial drone crops.

Drone imagery (the real target)

- Format: YOLO-labeled frames, 4096x3072 (12 MP), DJI camera (24 mm-equiv).
- Latest dataset (D:/data/invasive): 1095 images = 655 with cogongrass boxes + 440 negative (no cogongrass). Single class 'cogon'.
- Combines TWO flights/dates: 2026-04-22 (262 frames) and 2026-06-06 (819 frames) + 14 misc.
- Imagery is a near-continuous grass carpet; SAM2 confirmed it has NO separable plant instances (segments the whole field as one object) - which is why object detection was abandoned.
- Resolution: at ~30 ft flight, ground sample distance ~3.35 mm/px (cogongrass blade ~2-4 px wide - marginal).

Tiling

Frames are cut into 512 px tiles (~1.7 m ground at 30 ft), each resized to 224 for the CNN. A green-index (ExG) filter drops sky / non-vegetation tiles. Tile labels are derived from the boxes: a tile is cogongrass if >=30% covered by a box. Negatives (440 frames) yield all-negative tiles.

3. Methodology

Pipeline (scripts)

Script	Role
prep_images.py	Import + (optionally) downscale frames; full-res supported (PREP_MAX=4096)
boxes_to_tiles.py	Cut tiles, ExG sky filter, derive tile labels from boxes
train_tiles.py / _da.py	Train tile classifier; _da adds CLAHE + domain-randomization aug + regularization
train_tiles_collection.py	Cross-collection split: train 0606, TEST held-out 0422
train_tiles_dino_spatial.py	DINOv2 patch-feature backbone + conv head
tta_eval.py	Test-time adaptation (AdaBN / TENT) evaluation
threshold_sweep.py	Recall/precision vs decision threshold
heatmap_infer.py	Deployable: tile -> AdaBN -> classify -> coverage heatmap
label_tiles.py	Interactive multi-species tile labeler

Evaluation protocol (critical)

The honest metric is CROSS-COLLECTION: train on one flight, test on an entirely held-out flight. A random tile/frame split inflated accuracy to ~0.97 because the dataset is visually homogeneous (few distinct scenes) - that 0.97 does NOT predict performance on a new field.

4. Model Architecture

Best deployable model: ResNet18 + DA + AdaBN

Input 224x224 RGB tile -> ResNet18 backbone (ImageNet-pretrained, BatchNorm throughout) -> head: Dropout(0.4) + Linear(512 -> 2). Trained with class-weighted / balanced loss, label smoothing (0.1), AdamW, heavy augmentation. The BatchNorm layers are what enable AdaBN: at inference, BN running statistics are recomputed on the NEW field's tiles (no labels), which cancels cross-collection covariate shift.

Flow: tile(512px) -> resize 224 -> [Conv/BN/ReLU x ResNet18] -> global avg pool -> Dropout -> Linear -> softmax -> P(cogongrass).

Alternative: DINOv2-spatial

Frozen DINOv2 ViT-S/14 -> 16x16x384 PATCH feature map -> conv head (Conv-BN-ReLU x2 -> global pool -> Dropout -> Linear). Uses spatial patch tokens (not the CLS vector - that mistake cost us 11 points). Competitive (~0.83) but did not beat the ResNet on our data.

Earlier approaches (superseded)

- Close-up ResNet50 classifier (web data) - wrong domain for drone.
- YOLOv8 patch detector - detection is ill-posed for continuous grass (no object instances).
- DINOv2 frozen + CLS linear probe - wrong way to use DINOv2 (0.717).

5. Results Across All Models

Headline models

Model / approach	Metric	Score	Notes
Close-up classifier (ResNet50)	test bal acc	0.919	web close-ups, wrong domain
YOLOv8 patch detector	test mAP50	0.39	detection ill-suited to grass
Tile clf v1 (old data)	bal acc	0.97	HOLLOW (negatives were sky)
Tile clf v2 (random split)	test bal acc	0.967	INFLATED by homogeneity

Cross-collection results (the honest numbers; train 0606, test held-out 0422)

Configuration	SOURCE	+ AdaBN
Baseline (blurry 1280, no DA)	0.804	-
DA: CLAHE + aug + reg (blurry)	0.817	-
Full-res 256 px, no CLAHE	0.793	-
Full-res 256 px, CLAHE	0.794	-
Full-res 512 px, no CLAHE	0.815	0.839
Full-res 512 px, CLAHE (best)	0.820	0.838
DINOv2 frozen, CLS-linear	0.717	-
DINOv2-spatial + conv head	0.801	0.828

AdaBN consistently adds ~+2 pts. TENT test-time adaptation COLLAPSED (~0.53). Best deployable = 512 px + AdaBN ~= 0.84. The ~0.82-0.84 ceiling holds across BOTH backbones and every knob.

Decision-threshold sweep (best model + AdaBN, held-out 0422)

Threshold	Recall	Precision	Cogongrass missed
0.50 (default)	0.87	0.65	13%
0.30	0.92	0.59	8%
0.20 (recommended)	0.94	0.55	6%
0.15	0.95	0.52	5%
0.10	0.96	0.48	4%

False negatives (missed cogongrass) are far costlier than false positives for an invasive species, so operate at a LOW threshold (~0.20): catch 94% of cogongrass.

6. Key Findings & Lessons

- DOMAIN SHIFT is the core problem. In-collection ~0.97 vs cross-collection ~0.84. Random splits lie; always evaluate train-one-collection / test-another.
- ARCHITECTURE IS TAPPED OUT. Augmentation, CLAHE, resolution, tile size, regularization, and two backbones (ResNet + DINOv2) all land at ~0.82-0.84.
- DATA DIVERSITY is the dominant lever. Only 2 collections, visually homogeneous.
- LABELS ARE NOISY. Isolated 'false positives' are visually indistinguishable from labeled cogongrass -> most are UNLABELED cogongrass. Measured precision (~0.55) is a pessimistic floor; true performance is likely better.
- RECALL is the stuck metric (~0.64-0.74 raw; ~0.87+ with AdaBN at 0.5; up to 0.94 at threshold 0.20). Lowering the threshold buys recall cheaply.
- AdaBN is the best single technique - free, deployment-realistic, stacks with everything.
- Bigger tiles (512, more context) beat smaller (256, finer texture) for this task.
- 512px tiles: false positives cluster near real cogongrass (61% vs 45% baseline) -> boundary/label effect.

7. All Possible Routes & Next Steps

A. Data (highest value - this is the real bottleneck)

- MORE COLLECTIONS / SITES: the single biggest lever. Each new field de-confounds the model.
- FLY LOWER (~20 ft) + higher-res sensor: cogongrass blade texture needs resolution (Purdue & BASF papers fly ~16-20 ft at 0.5-1.7 mm/px vs our 3.35).
- MULTI-TEMPORAL RGB (phenology): fly the SAME fields 2-3x across the season. Cogongrass has distinctive timing (early green-up, persistent thatch, reddish senescence). Literature: beats adding spectral bands for telling similar grasses apart. Uses existing camera.
- MODEL-ASSISTED LABEL CLEANING: pre-fill tiles with model predictions, human corrects. Fixes the label gaps inflating false positives; gives a TRUE performance number.

B. Sensors

- MULTISPECTRAL (NIR + red-edge): NDVI for vegetation-soil separation (the #1 cross-domain failure mode), NDRE/red-edge for grass-vs-grass. More illumination-invariant -> helps domain shift. Combine via channel-stacking (RGB+NIR+indices); adapt the first conv layer. Needs a multispectral sensor + reflight + band registration; store tiles as TIFF/npy.

C. Model / algorithm (lower priority - architecture is tapped out)

- DINOv2 SEGMENTATION with a dense decoder (needs pixel labels) - the BASF SOTA for ground->aerial shift.
- Domain-generalization aug: MixStyle, Fourier/amplitude-mix (style) augmentation.
- IBN-Net (InstanceNorm) for style-invariant features; self-supervised pretrain on unlabeled drone frames.
- Fix TENT (lower lr / class-balanced entropy) to stack on AdaBN.

D. Inference / deployment

- Operate at threshold ~0.20 (prioritize recall - don't miss the invasive).
- Merge adjacent positive tiles into stands -> boundary false-positives vanish at field level.
- Overlapping tiles at inference + hierarchical fallback (flag low-confidence tiles for human review).

8. Recommended Path Forward

The model is in good shape and likely better than its metrics (label noise). Effort should now go to DATA and LABELS, not architecture:

Priority	Action	Why
1	Collect 1-2 more sites/collections	De-confounds; the proven lever
2	Model-assisted label cleaning pass	FPs are mostly unlabeled cogongrass; get true score
3	Fly ~20 ft + multi-temporal (phenology)	Cheapest capture change; hits grass-vs-grass
4	Deploy ResNet-DA + AdaBN at thr 0.20	Usable now; high recall
5	Add multispectral (NIR/red-edge)	Veg-soil + domain robustness; needs hardware

Current deliverable: a validated ~0.84 cross-collection model with an AdaBN inference tool (heatmap_infer.py) and a full, reproducible pipeline. The project went from 'is this possible?' to a rigorously-measured, deployable system that knows exactly what it needs next.